



**Sanjivani Rural Education Society's**  
**Sanjivani College of Engineering, Kopargaon-423 603**  
*(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)*  
**NAAC 'A' Grade Accredited, ISO 9001:2015 Certified**

## **Department of Computer Engineering**

*(NBA Accredited)*

**Subject- Object Oriented Programming (CO212)**  
**Unit 2 – Overloading and Inheritance**  
**Topic – 2.8 Constructor and Destructor in Inheritance**

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# Constructor and Destructor in Inheritance

- Invocation of constructors and destructors depends on the type of inheritance being implemented.

## Constructor and destructor in single inheritance:

- Base class constructors are called first and the derived class constructors are called next in single inheritance.
- Destructor is called in reverse sequence of constructor invocation i.e. The destructor of the derived class is called first and the destructor of the base class is called next.



# Example of Constructor and Destructor in Single Inheritance

|                                                                                                                                                                                                                            |                                                                                                                                                                                                    |                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>class base {     public:     base()     {         cout&lt;&lt;"base class constructor";     }     ~base()     {         cout&lt;&lt;"base class destructor";     } };  class derived: public base {     public:</pre> | <pre>    derived()     {         cout&lt;&lt;"derived class constructor";     }     ~derived()     {         cout&lt;&lt;"derived class destructor";     } };  int main() {     derived d; }</pre> | <p><b>Output :</b></p> <p>base class constructor<br/>derived class constructor<br/>derived class destructor<br/>base class destructor</p> |
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# Constructor and destructor in multiple inheritance

- Constructors from all base classes are invoked first and then derived class constructor is called.
- Order of constructor invocation depends on the order of how base classes are inherited.
- For example:
- `class D:public B, public C { //... }`
- Here, B is inherited first, so the constructor of class B is called first and then constructor of class C is called next.
- However, the destructor of derived class is called first and then destructor of the base class which is mentioned in the derived class declaration is called from last towards first in sequentially.



# Example of constructor and destructor in multiple inheritance

```
class base_one
{
public:
    base_one()
    {
        cout<<"base_one class constructor";
    }
    ~base_one()
    {
        cout<<"base_one class destructor";
    }
};

class base_two
{
public:
    base_two()
    {
        cout<<"base_two class constructor";
```

```
    }
    ~base_two()
    {
        cout<<"base_two class destructor";
    }
};

class derived: public base_one, public
    base_two
{
public:
    derived()
    {
        cout<<"derived class constructor";
    }
    ~derived()
    {
        cout<<"derived class destructor";
    }
};
```

```
};

int main()
{
    derived d;
    return 0;
}

Output :
```

```
base_one class constructor
base_two class constructor
derived class constructor
derived class destructor
base_two class destructor
base_one class destructor
```

Thank you..